

A Proof of the Snake Lemma (For R -Modules)

Grant Talbert
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Boston University

1 Proof

Lemma 1 (The Snake Lemma). *Let the diagram of R -modules*

$$\begin{array}{ccccccccc} 0 & \longrightarrow & L_1 & \xrightarrow{\alpha_1} & M_1 & \xrightarrow{\beta_1} & N_1 & \longrightarrow & 0 \\ & & \downarrow \lambda & & \downarrow \mu & & \downarrow \nu & & \\ 0 & \longrightarrow & L_0 & \xrightarrow{\alpha_0} & M_0 & \xrightarrow{\beta_0} & N_1 & \longrightarrow & 0 \end{array}$$

commute, where rows are exact complexes. Then there exists an exact sequence

$$0 \longrightarrow \ker \lambda \longrightarrow \ker \mu \longrightarrow \ker \nu \xrightarrow{\delta} \operatorname{coker} \lambda \longrightarrow \operatorname{coker} \mu \longrightarrow \operatorname{coker} \nu \longrightarrow 0.$$

Proof. For convenience of notation, we will denote rows by $1_\bullet$ and $0_\bullet$, and columns by $L_\bullet$, $M_\bullet$, and $N_\bullet$.

Clearly, there exists the morphism $0 : 0 \rightarrow \ker \lambda$ with $\ker 0 = 0$ and $\operatorname{im} 0 = \{0\}$.

Since $1_\bullet$ is exact, $\ker \alpha_1 = \operatorname{im} 0 = \{0\}$. Consider the restriction $\alpha_1|_{\ker \lambda}$. Note that for any $a \in \ker \lambda$,

$$\begin{array}{ccc} a & \xrightarrow{\alpha_1} & \alpha_1(a) \\ \lambda \downarrow & & \downarrow \mu \\ 0 & \longrightarrow & 0 \end{array}$$

because the diagram commutes. We must then have $\operatorname{im} \alpha_1|_{\ker \lambda} \subseteq \ker \mu$. Therefore, the complex

$$0 \xrightarrow{0} \ker \lambda \xrightarrow{\alpha_1|_{\ker \lambda}} \ker \mu$$

is exact at $0, \ker \lambda$.

Now consider the restriction $\beta_1|_{\ker \mu} : \ker \mu \rightarrow N_1$. First, we will show $\operatorname{im} \beta_1 \subseteq \ker \nu$. Let $a \in \ker \mu$, and note that

$$\begin{array}{ccc} a & \xrightarrow{\beta_1} & \beta_1(a) \\ \mu \downarrow & & \downarrow \nu \\ 0 & \longrightarrow & 0 \end{array}$$

because the diagram commutes. Therefore, $\beta_1|_{\ker \mu} : \ker \mu \rightarrow \ker \nu$. Also note that because $1_\bullet$ is exact, we have $a \in \operatorname{im} \alpha_1 \iff a \in \ker \beta_1$. Restricting these sets to the subset $\ker \mu$ does not change this fact, so we have that the complex

$$0 \xrightarrow{0} \ker \lambda \xrightarrow{\alpha_1|_{\ker \lambda}} \ker \mu \xrightarrow{\beta_1|_{\ker \mu}} \ker \nu$$

is exact at $\ker \mu$.

Now we must show the morphism $\delta : \ker \nu \rightarrow \operatorname{coker} \lambda$ exists, which will be done by a diagram

chase. Consider the diagram

$$\begin{array}{ccccccc}
 & 0 & & 0 & & 0 & \\
 & \downarrow & & \downarrow & & \downarrow & \\
 0 & \longrightarrow & \ker \lambda & \longrightarrow & \ker \mu & \longrightarrow & \ker \nu \\
 & \downarrow & & \downarrow & & \downarrow & \\
 0 & \longrightarrow & L_1 & \xrightarrow{\alpha_1} & M_1 & \xrightarrow{\beta_1} & N_1 \longrightarrow 0 \\
 & \downarrow \lambda & & \downarrow \mu & & \downarrow \nu & \\
 0 & \longrightarrow & L_0 & \xrightarrow{\alpha_0} & M_0 & \xrightarrow{\beta_0} & N_0 \longrightarrow 0 \\
 & \downarrow & & \downarrow & & \downarrow & \\
 & \text{coker } \lambda & \longrightarrow & \text{coker } \mu & \longrightarrow & \text{coker } \nu & \longrightarrow 0 \\
 & \downarrow & & \downarrow & & \downarrow & \\
 & 0 & & 0 & & 0 &
 \end{array} , \tag{1}$$

where all unexplained arrows are induced completely naturally. Let $a \in \ker \nu$, and consider the chase

$$\begin{array}{ccccccc}
 & 0 & & 0 & & 0 & \\
 & \vdots & & \vdots & & \vdots & \\
 0 & \cdots & & & & a & \\
 & \vdots & & \vdots & & \downarrow & \\
 0 & \cdots & & c & \xrightarrow{\beta_1} & b & \cdots 0 \\
 & \vdots & & \downarrow \mu & & \downarrow \nu & \\
 0 & \cdots & e & \xrightarrow{\alpha_0} & d & \xrightarrow{\beta_0} & * \cdots 0 \\
 & \vdots & \downarrow & & \vdots & & \vdots \\
 & & f & & & & 0 \\
 & \vdots & & \vdots & & \vdots & \\
 & 0 & & 0 & & 0 &
 \end{array} . \tag{2}$$

We will show this diagram induces a homomorphism $\delta : \ker \nu \rightarrow \text{coker } \lambda$. The map $\ker \nu \hookrightarrow N_1$ is the inclusion map, the map $M_1 \rightarrow M_0$ is μ , and the map $L_0 \twoheadrightarrow \text{coker } \lambda$ is the natural projection. The only maps that need sufficient explaining are the maps $N_1 \rightarrow M_1$ and $M_0 \rightarrow L_0$.

Given that $b = \iota(a)$ for $\iota : \ker \lambda \hookrightarrow N_1$ the inclusion map, we define c as the preimage $c := \beta_1^{-1}(b)$. Such a c must exist, as β_1 must be epic for $1_\bullet$ to be exact by lemma 2. The question is whether this operation is well-defined; that is, different choices $c_1, c_2 \mapsto b$ must map to the same end result if we continue along the sequence. To show this, we must first define the map $L_0 \rightarrow M_0$.

This map is defined similarly. Since the diagram commutes, $\beta_0(d) = \nu(b)$. However, since $b \in \ker \nu$, we have $\beta_0(d) = 0$. It follows that $d \in \ker \beta$, and since $0_\bullet$ is exact, we have that $\text{im } \alpha_0 = \ker \beta_0$, and there exists an $e \in L_0$ such that $\alpha_0(e) = d$. Additionally, note that since $0_\bullet$ is exact, α_0 is mono by lemma 2, so this e is unique. Therefore, we can define the map $M_0 \rightarrow L_0$ such that $d \mapsto \alpha_0^{-1}(d)$, and this is well-defined.

Now to show the map $b \mapsto c$ is well-defined. Let $c_1, c_2 \in M_1$ such that $\beta_1(c_1) = \beta_1(c_2) = b$. Since β_1 is a homomorphism, $\beta_1(c_1 - c_2) = 0$. Therefore, $c_1 - c_2 \in \ker \beta_1$. Since $1_\bullet$ is exact, $c_1 - c_2 \in \text{im } \alpha_1$. Since $1_\bullet$ is exact, α_1 is mono by lemma 2, so there exists precisely one $g \in L_1$ such that $\alpha_1(g) = c_1 - c_2$. Consider the diagram

$$\begin{array}{ccccccc} 0 & \cdots & g & \xrightarrow{\alpha_1} & (c_1 - c_2) & \xrightarrow{\beta_1} & 0 \cdots 0 \\ & & \downarrow \lambda & & \downarrow \mu & & \\ 0 & \cdots & \lambda(g) & \cdots & & & \\ & & \downarrow & & & & \\ & & 0 & \cdots & & & \end{array} .$$

We need only explain $\lambda(g) \rightarrow 0$. It suffices to show columns are exact in diagram (1), since this would imply $\text{im } \lambda = \ker \pi$ for $\pi : L_0 \rightarrow \text{coker } \lambda$ the canonical projection. It suffices to show much less, actually, but I will just show this broader statement. Consider a complex of the form

$$0 \longrightarrow \ker \lambda \xhookrightarrow{\iota} L_1 \xrightarrow{\lambda} L_0 \xrightarrow{\pi} \text{coker } \lambda \longrightarrow 0$$

Since $\iota : \ker \lambda \hookrightarrow L_1$ is injective by the universal property of kernels, we have that $\ker \iota = \{0\} = \text{im } 0$, and the complex is exact at $\ker \lambda$. Note that $\ker \lambda = \iota(\ker \lambda) = \text{im } \iota$, so the complex is exact at L_1 . Now note that since $\text{coker } \lambda \cong L_0 / \text{im } \lambda$, we have that $\text{im } \lambda \subseteq \ker \pi$. Additionally, $x \in \ker \pi$ implies that $\pi(x) = x + \text{im } \lambda = \text{im } \lambda$. By properties of cosets, it follows that $x \in \text{im } \lambda$. Therefore, the complex is exact at L_0 . Finally, recall the universal property of cokernels: the cokernel $\text{coker } \lambda$ must be initial with respect to morphisms $\beta : L_0 \rightarrow Z$ such that the diagram

$$\begin{array}{ccccc} & & 0 & & \\ & \curvearrowright & & \curvearrowright & \\ L_1 & \xrightarrow{\lambda} & L_0 & \xrightarrow{\beta} & Z \\ & & \downarrow \pi & \nearrow \bar{\beta} & \\ & & \text{coker } \lambda & & \end{array}$$

commutes. Note that π is epic, so $\text{im } \pi = \text{coker } \lambda = \ker 0$, so the complex is exact at $\text{coker } \lambda$. Rows are thus exact complexes, and we have that $\pi(\lambda(g)) = 0$ since $\text{im } \lambda = \ker \pi$.

Now back to showing $b \mapsto c$ is well-defined. Suppose that $c_2 \mapsto d \mapsto e$. Since the diagram commutes, we must have $\mu\alpha_1(g) = \alpha_0\lambda(g)$. We also have that $c_1 = c_2 + (c_1 - c_2) = c_2 + \alpha_1(g)$. Since $0_\bullet$ is epic, by lemma 2, α_0 is mono, and thus the only thing mapping to d is e , and thus $\mu(c_2) = \alpha_0(e)$ by commutativity. Now note that by commutativity of the diagram, $\alpha_0(e + \lambda(g)) = \mu(c_1) = \mu(c_2 + c_1 - c_2)$:

$$\begin{array}{ccc} & c_2 + c_1 - c_2 & \\ & \downarrow & \\ e + \lambda(g) & \longrightarrow & d + \alpha_0(\lambda(g)) \end{array}$$

Since $\lambda(g)$ dies in $\text{coker } \lambda$, we have that $c_1 \mapsto f + 0 = f$ in diagram (2). Therefore, c_1, c_2 eventually map to the same thing, and δ is well-defined overall as the map

$$a \xrightarrow{\iota} b \xrightarrow{\beta_1^{-1}} c \xrightarrow{\mu} d \xrightarrow{\alpha_0^{-1}} e \xrightarrow{\pi} f.$$

Now, we must show the complex

$$0 \xrightarrow{0} \ker \lambda \xrightarrow{\alpha_1|_{\ker \lambda}} \ker \mu \xrightarrow{\beta_1|_{\ker \mu}} \ker \nu \xrightarrow{\delta} \operatorname{coker} \lambda$$

is exact at $\ker \nu$. First, we show $\operatorname{im} \beta_1|_{\ker \mu} \subseteq \ker \delta$. Consider the commutative diagram

$$\begin{array}{ccc} \ker \mu & \xrightarrow{\beta_1} & \ker \nu \\ \downarrow \iota & & \downarrow \iota \\ M_1 & \xleftarrow{\beta_1^{-1}} & N_1 \\ \downarrow \mu & & \\ 0 & & \end{array}$$

Notice that by the commutativity of this diagram, we have that

$$a \xrightarrow{\beta_1} \beta_1(a) \xrightarrow{\iota} \beta_1(a) \xrightarrow{\beta_1^{-1}} a \xrightarrow{\mu} 0 \xrightarrow{\pi \alpha_0^{-1}} 0$$

Therefore, $\operatorname{im} \beta_1 \subseteq \ker \delta$. Now we show the converse. Let $a \in \ker \delta$. It follows that at some point in the sequence

$$\delta : a \xrightarrow{\iota} b \xrightarrow{\beta_1^{-1}} c \xrightarrow{\mu} d \xrightarrow{\alpha_0^{-1}} e \xrightarrow{\pi} f,$$

at least one of these elements must be 0. Also note that if one element is 0, all subsequent elements must also be 0. We will go through this one at a time.

If $a = 0$, then $a \in \operatorname{im} \beta_1|_{\ker \mu}$ because $0 \mapsto 0$ and $0 \in \ker \mu$. If $b = 0$, then $a = 0$ because the inclusion ι is injective. If $c = 0$, then $\beta_1(0) = b$ implies that $b = 0$. If $d = 0$, then $\mu(c) = 0$, and thus $c \in \ker \mu$. It follows that $\beta_1(c) \in \operatorname{im} \beta_1|_{\ker \mu}$. Note that

$$\begin{array}{ccc} \ker \mu & \longrightarrow & \ker \nu \\ \downarrow & & \downarrow \\ M_1 & \longrightarrow & M_0 \end{array}$$

commutes, and thus $\beta_1(c) = a$. Therefore, $a \in \operatorname{im} \beta_1|_{\ker \mu}$. If $e = 0$, then because α_0 is injective, $d = 0$, and $a \in \operatorname{im} \beta_1|_{\ker \mu}$.

If $f = 0$, then note that

$$\begin{array}{ccccc} L_1 & \xrightarrow{\alpha_1} & M_1 & \xrightarrow{\beta_1} & N_1 \\ \downarrow \lambda & & \downarrow \mu & & \downarrow \nu \\ \operatorname{im} \lambda & \xrightarrow{\alpha_0} & \operatorname{im} \mu & \xrightarrow{\beta_0} & \operatorname{im} \nu \end{array}$$

commutes. Note that $f = 0$ if and only if $e \in \operatorname{im} \lambda$, and we have that $\beta_0 \alpha_0(\lambda) = 0$ because rows are exact. Then $\nu(b) = \beta_0 \alpha_0(\lambda)$, and thus $b \in \ker \nu$. Great, we already knew that. Note that e has a preimage under λ , which must map to c by commutativity of the diagram. Since rows are exact, $\beta_1(c) = 0$, and thus $b = 0$. Therefore, $\ker \delta \subseteq \operatorname{im} \beta_1|_{\ker \mu}$, and the complex is exact at $\ker \nu$. ■

A Definitions and Results Used in the Proof

Lemma 2. *Let*

$$0 \longrightarrow M_2 \xrightarrow{d_1} M_1 \xrightarrow{d_0} M_0 \longrightarrow 0$$

be a short exact chain complex. Then d_1 is a monomorphism and d_0 is an epimorphism.

Definition 1 (Kernel). The kernel $\ker \phi$ of an R -module homomorphism $\phi : R \rightarrow S$ is the R -module which is final with respect to mappings $\alpha : S \rightarrow Z$ such that $\phi\alpha = 0$. That is, $\ker \phi$ is the R -module such that

$$\begin{array}{ccccc} & & 0 & & \\ & \searrow & \curvearrowright & \searrow & \\ R & \xrightarrow{\phi} & S & \xrightarrow{\alpha} & Z \\ & \swarrow \text{dotted} & \uparrow & & \\ & & \ker \phi & & \end{array}$$

commutes.